

Pneumococcus and Immunosenescence

Susceptibility to pneumonia and bacterial and viral infections (MBL2, FCGR2A, TIRAP, TLR4), pneumonia scores, zone, otitis, severe urinary infection and COVID-19, and what this implies for pneumococcal, flu and zone vaccines

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Genetic **MBL** is preserved: alleles B and C of MBL2 are absent and there is only 1 allele D (genotype A/O, intermediate MBL, not deficient). TLR4 Asp299Gly is absent. The severe COVID-19 is in percentile 4.7 and the sepsis in the 3.2 loci (with few loci; in §44 the sepsis score was in the 23rd percentile). Low respiratory infection is typical (45.2). In §44, the base immunity is favourable: Low genetic CRP (the 12th percentile), neutrophils in 20, lymphocytes in 70 and genetic vitamin D in 91.

Unfavourable. The genomic score of pneumonia is in percentile 84.2 (typical loci load, 59.6). Otitis media is in 85.3 and urinary infection in 93.8. FCGR2A H131R is heterozygous (1 allele Arg131, which binds worse to IgG2 against the pneumococcal capsule). The herpes zóster (zone) is in percentile 74.9 and the cellulite in the 66.8.

What does it mean? Innate defense (MBL, TLR4) is good, but there is a polygenic tendency for bacterial mucosal infections (pneumonia, otitis, urinary infection). This is consistent with FCGR2A H/R, which makes the IgG2 opsonization less efficient. The most useful is to have the vaccines: pneumococcal conjugated, annual flu, recombinant zone and COVID-19/RSV according to age. Dapagliflozin increases urinary and genital infections, and the urinary infection score is high.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 6 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 6 of the chip, 0 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 7 PGS Catalog scores calculated locally with 3.784.593 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 32 loci, 18 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a higher percentile = more susceptibility. The scores of low respiratory infection, zone and cellulitis are those of §44.

Pneumonia and respiratory tract

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Pneumon	PGS019283 (Chen 2025, PRS-CS, 984 thousand variants)	950.710 983.820 (weighted coverage 96%)	84,2	MAIS PROVABLE	unfavourable
Acute low respiratory infection	PGS000925 (Tanigawa 2022, 400 variants)	339 400 (weighted coverage 84%)	45,2	TIPIC	typical
Suppurative otitis media	PGS019079 (Chen 2025, PRS-CS, 984 thousand variants)	950.719 983.826 (weighted coverage 96%)	85,3	MAIS PROVABLE	unfavourable
COVID-19 grave	PGS004938 (Kovalenko 2024, LDpred2, 956 thousand variants)	931.920 955.503 (weighted coverage 97%)	4,7	MENOS PROVABLE	favourable

- **Acute low respiratory infection:** Já no §44.

Other infections

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Herpes zóster (zone)	PGS001131 (Tanigawa 2022, 82 variants)	62 81 (weighted coverage 66%)	74,9	TIPIC	typical
Urinary infection	PGS019461 (Chen 2025, PRS-CS, 984 thousand variants)	950.715 983.825 (weighted coverage 96%)	93,8	MAIS PROVABLE	unfavourable
Cellulitis and abscesses	PGS001869 (Tanigawa 2022, 131 variants)	128 131 (weighted coverage 99%)	66,8	TIPIC	typical

- **Herpes zóster (zone):** Já no §44.
- **Cellulitis and abscesses:** Já no §44.

4. Key variants by topic

Variants of innate immunity with known effect on pneumococcal and bacterial infection.

Recognition of pneumococci and other bacteria

The mannose binding lectin (MBL) recognizes the sugars of the bacteria surface (including pneumococci) and activates the complement. The B, C and D alleles of MBL2 lower the MBL, especially in combination (O/O genotypes).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MBL2	rs1800450	codon 54 (B)	C/C	chip	T (0 copy/s)	favourable	Codon 54 (B allele, Gly54Asp): low mannose binding lectin; in combination with another deficient allele, plus pneumonia and invasive pneumococcal infection	Moderate
MBL2	rs1800451	codon 57 (C)	C/C	chip	T (0 copy/s)	favourable	Codon 57 (allele C, Gly57Glu): low MBL	Moderate
MBL2	rs5030737	codon 52 (D)	G/A	chip	A (1 copy/s)	unfavourable (light)	Codon 52 (D allele, Arg52Cys): low MBL	Moderate

Innate response and opsonization

The FcγRIIIa receptor (FCGR2A) allows phagocytes to swallow bacteria covered by IgG2, the class of antibodies that pneumococcal vaccine induces. TIRAP and TLR4 regulate the innate response to bacteria.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FCGR2A	rs1801274	H131R	A/G	chip	G (1 copy/s)	unfavourable (light)	H131R: Arg131 (G) binds poorly to IgG2, the class of antibodies against the pneumococcal capsule; more pneumococcal bacteremia in homozygous R/R	Moderate
TIRAP (Mal)	rs8177374	S180L	C/C	chip	T (0 copy/s)	without judgment	S180L: heterozygous have a balanced TLR2/TLR4 response and less invasive pneumococcal disease in some studies; homozygous without protection	Weak
TLR4	rs4986790	D299G	A/A	chip	G (0 copy/s)	favourable	Asp299Gly: reduced response to LPS; associated with more severe bacterial infections in some studies	Weak

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 5 \times 10^{-8}$) of pneumonia and respiratory and sepsis infections. Alleles are oriented towards more risk and the sum is compared with the reference population. For influenza and the response to vaccines there are too few loci usable.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Pneumonia and respiratory infections	24 (16)	13 de 32	16	59,6	typical
Sepsis	7 (2)	0 de 4	2	3,2	favourable

6. Favourable

- **MBL2 without alleles B and C** (54 and 57 C/C codons); 1 allele D (codon 52 G/A): genotype A/O, MBL intermediate, without deficiency. **Moderate.**
- **TLR4 Asp299Gly A/A**: normal response to LPS. **Weak.**
- **COVID-19 grave** in percentile 4.7. **Moderate.**
- **Sepsis**: loci load in 3.2 (only 2 loci usable; §44 score in the 23rd percentile). **Weak.**
- **Acute low respiratory infection** in percentile 45.2; load of pneumonia loci and respiratory infections in 59.6. **Weak moderated.**
- **Base immunity (§44)**: CRP genetics in the 12th percentile, neutrophils in the 20, vitamin D genetics in the 91. **Moderate.**

7. Unfavourable

- **Pneumon** in percentile 84.2 (genomic score). **Moderate.**
- **Urinary infection** in percentile 93.8; the effect of dapagliflozin (glucose in urine) is added. **Moderate.**
- **Suppurative otitis media** in percentile 85.3. **Weak moderated.**
- **FCGR2A H131R A/G**: 1 Arg131 allele (less efficient IgG2 opsonization; the highest risk is in R/R). **Weak moderated.**
- **Herpes zoster** in percentile 74.9 and cellulite in the 66.8 (typical-high). **Weak.**
- **TIRAP S180L C/C**: without heterozygosity associated with less invasive pneumococcal disease (most people are like this). **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Immunosenescence	With age, the response to polysaccharide vaccines and cellular immunity against the zone decrease; pneumonia in percentile 84.2 and zone in the 74.9	Unfavourable (mild)
Inflammaging	Low CRP (§44) and severe COVID-19 in percentile 4.7	Favourable
Chronic Inflammation	Repeated infections (urine, otitis, pneumonia) add inflammatory episodes	Unfavourable (conditional)
Intercellular Communication	IgG2 (FCGR2A H/R) and complement via MBL (A/O)	Mixed
Loss of T Naive Cells	Thymus and naive T cells decrease with age; the recombinant vaccine in the zone compensates for loss of specific immunity	Conditional
Intestinal Barreira dysfunction	Mucosal barriers (respiratory, urinary and intestinal) are the first line against bacteria	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Conjugated pneumococcal vaccine:** a dose of PCV20 or PCV21. It is recommended from the 65 years and in risk groups; with the high pneumonia score and the use of rapamycin, it is reasonable to discuss it with the doctor already (without reinforcement for most people). *Suggestion (not applied).*
2. **Influenza every year,** in October-November (now). From the 65 years, high dose vaccine or adjuvant. *Suggestion (not applied).*
3. **Zone (herpes zoster):** recombinant vaccine with adjuvant, 2 doses with 2-6 months apart, from the 50 years (or earlier if there is immunosuppression). It is not a live vaccine, so it is compatible with rapamycin. *Suggestion (not applied).*
4. **COVID-19** according to the annual campaign, RSV from the 60 years (shared decision) and dTpa every 10 years. *Suggestion (not applied).*
5. **Rapamycin and vaccines:** to give the vaccines in the middle of the 6 cycle weeks, far from taking 4 mg (at least ~2 weeks later). In the elderly, intermittent inhibition of mTOR improved the response to influenza vaccine, but taking close to the vaccine can reduce it. *Suggestion (not applied).*
6. **Urinary and genital infections with dapagliflozin:** drinking enough water, daily genital hygiene and urinating after relationships. Seeking the doctor early if there is burning, fever or low back pain. *Suggestion (not applied).*
7. **Pneumon:** do not ignore fever with cough and shortness of breath; treat early. Hand hygiene and mask in times of flu and closed places reduce respiratory infections. *Suggestion (not applied).*
8. **In the plane and useful for immunity:** vitamin D 10 000 IU, zinc 20-25 mg with copper, sleep of 7-9 hours, regular exercise and zero tobacco. *Suggestion (not applied).*
9. **Oral health** (oral health report): plaque and periodontitis are sources of aspirated bacteria and are associated with pneumonia. *Suggestion (not applied).*

10. Limitations

- There are no validated scores for vaccine response; the response depends mainly on age, immunosuppression and type of vaccine.
- The scores of pneumonia, otitis and urinary infection (Chen 2025) are based on diagnoses of health records; urinary infection is greatly influenced by sex and anatomy.
- MBL2 was read only in the 3 structural variants; the promoter variants (H/L, X/Y) were not evaluated.
- The flu loci, sepsis and antibodies against pneumococci are few or without published risk allele.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palindromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (7 scores: pneumonia, low respiratory infection, otitis media, severe COVID-19, zone, urinary infection and cellulitis; 3 already used in §44). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510-8$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** Roy 2002 and Eisen 2008 (MBL2 and pneumococci), Yee 2000 and Endeman 2009 (FCGR2A), Khor 2007 (TIRAP), Mannick 2014 (mTOR inhibition and influenza vaccine), adult vaccination guidelines (PCV20/PCV21, recombinant zone vaccine, flu and RSV), GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Recognition of pneumococci and other bacteria	MBL2	rs1800450	C/C	chip	T	0	strand +	without the risk allele
Recognition of pneumococci and other bacteria	MBL2	rs1800451	C/C	chip	T	0	strand +	without the risk allele
Recognition of pneumococci and other bacteria	MBL2	rs5030737	G/A	chip	A	1	strand +	1 copy of risk allele
Innate response and opsonization	FCGR2A	rs1801274	A/G	chip	G	1	strand +	1 copy of risk allele
Innate response and opsonization	TIRAP (Mal)	rs8177374	C/C	chip	T	0	strand +	0 copy of the effect allele
Innate response and opsonization	TLR4	rs4986790	A/A	chip	G	0	strand +	without the risk allele